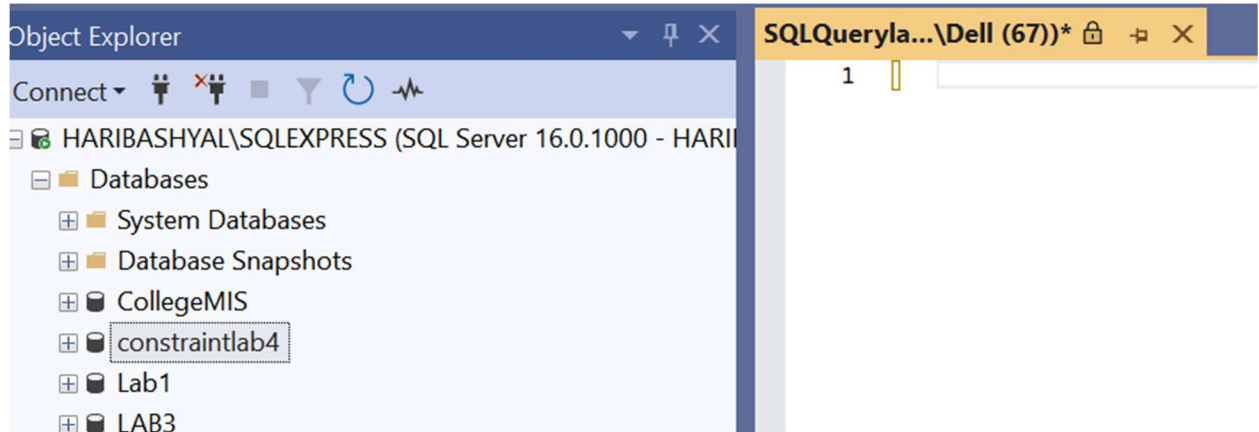


LAB-4: Use of Constraints

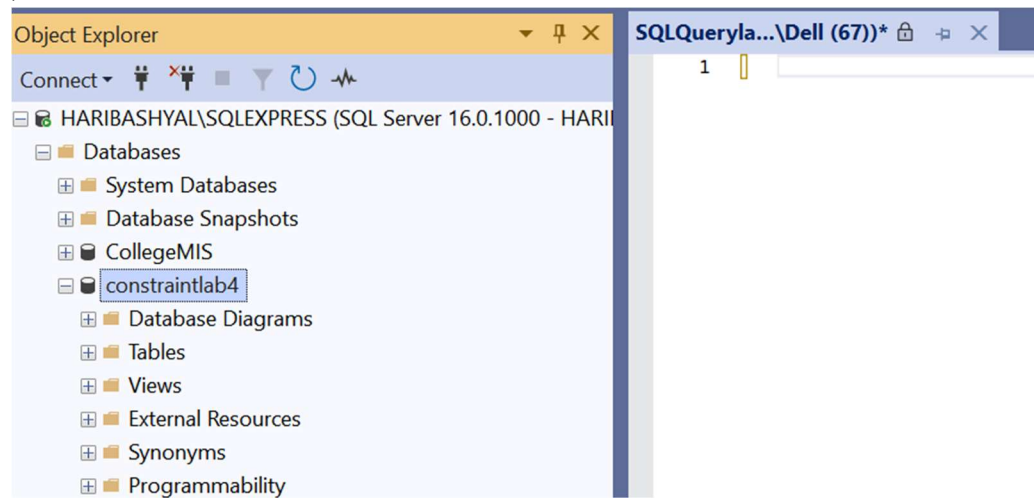
A. Creating Database:

Query: CREATE DATABASE constraintlab4



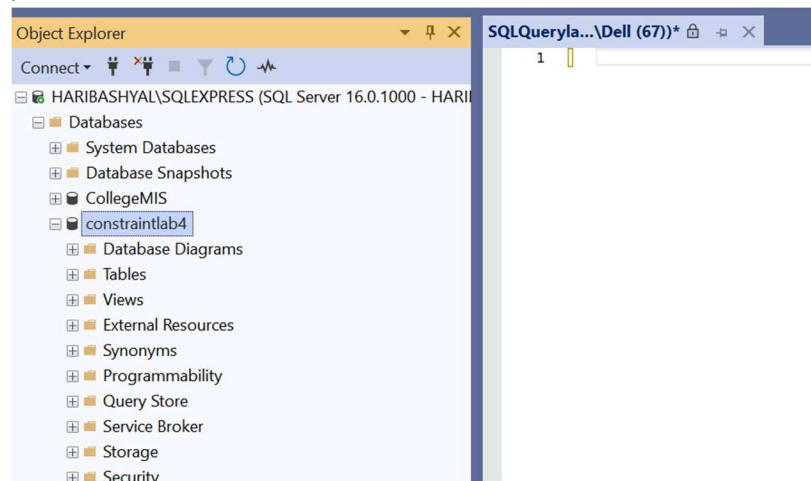
B. Creating Table of Department:

Query: CREATE TABLE department(
 did int primary key,
 dname varchar(20) NOT NULL,
 location varchar(20) default 'head office'
)



C. Creating Table of Employee:

Query: CREATE TABLE employee(
 eid int identity(1,1) primary key,
 fname varchar(20) NOT NULL,
 mname varchar(20),
 lname varchar(20) NOT NULL,
 Email varchar(20) unique,
 salary decimal(10,2) check(salary >= 18000),
 did int , foreign key(did) references department(did),
 joindate date default getdate()
)



1. Insert sample data into all tables (Department, Employee).

Query: INSERT INTO department (did, dname, location) VALUES
 (1, 'Human Resources', 'Kathmandu'),
 (2, 'Finance', 'Pokhara'),
 (3, 'IT', 'Lalitpur'),
 (4, 'Marketing', 'Biratnagar'),
 (5, 'Sales', 'Butwal'),
 (6, 'Operations', 'Dharan'),
 (7, 'Legal', 'Bhaktapur'),
 (8, 'Research', 'Janakpur'),
 (9, 'Customer Service', 'Nepalgunj'),
 (10, 'Administration', 'head office');

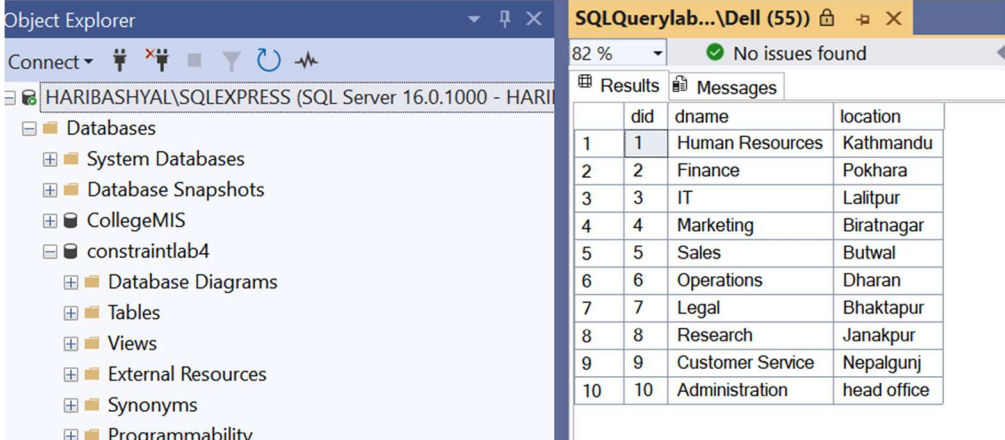
```

INSERT INTO employee (fname, mname, lname, Email, salary, did, joindate)
VALUES
('Rajan', 'Kumar', 'Joshi', 'rajan.joshi@ex.com', 24000.00, 1, '2024-05-10'),
('Sneha', NULL, 'Tamang', 'sneha.tamang@ex.com', 22000.00, 2, '2023-10-01'),
('Bibek', 'Prasad', 'Shah', 'bibek.shah@ex.com', 26000.00, 3, '2022-08-15'),
('Laxmi', NULL, 'Malla', 'laxmi.malla@ex.com', 20000.00, 4, '2025-02-20'),
('Kamal', 'Raj', 'Thapa', 'kamal.thapa@ex.com', 23000.00, 3, '2023-06-30'),
('Nabin', NULL, 'Rai', 'nabin.ra123@ex.com', 21000.00, 6, '2024-07-12'),
('Pooja', 'Devi', 'Shrestha', 'pooja.shres@ex.com', 25000.00, 1, '2022-11-05'),
('Suman', NULL, 'Lama', 'suman.lama12@ex.com', 19500.00, 8, '2023-03-18'),
('Asha', 'Kumari', 'Yadav', 'asha.yadav12@ex.com', 21500.00, 9, '2024-09-25'),
('Manoj', NULL, 'Karki', 'manoj.karki@ex.com', 27000.00, 2, '2025-01-01');

```

2. Display all Department.

Query: select*
from department

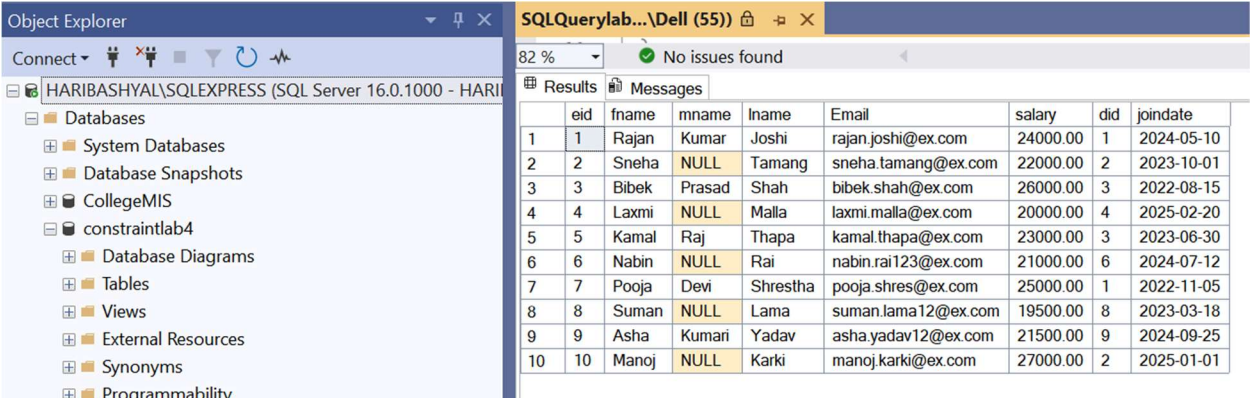


The screenshot shows the SQL Server Enterprise Manager interface. The 'Object Explorer' on the left displays the database structure for 'HARIBASHYAL\SQLEXPRESS (SQL Server 16.0.1000 - HARIBASHYAL)'. The 'Results' tab on the right shows the output of the query 'select* from department'. The table has 10 rows and 4 columns: 'did', 'dname', and 'location'.

did	dname	location
1	Human Resources	Kathmandu
2	Finance	Pokhara
3	IT	Lalitpur
4	Marketing	Biratnagar
5	Sales	Butwal
6	Operations	Dharan
7	Legal	Bhaktapur
8	Research	Janakpur
9	Customer Service	Nepalgunj
10	Administration	head office

3. Show all Employee.

Query: select*
from Employee



The screenshot shows the SQL Server Enterprise Manager interface. The 'Object Explorer' on the left displays the database structure for 'HARIBASHYAL\SQLEXPRESS (SQL Server 16.0.1000 - HARIBASHYAL)'. The 'Results' tab on the right shows the output of the query 'select* from Employee'. The table has 10 rows and 9 columns: 'eid', 'fname', 'mname', 'lname', 'Email', 'salary', 'did', and 'joindate'.

eid	fname	mname	lname	Email	salary	did	joindate
1	Rajan	Kumar	Joshi	rajan.joshi@ex.com	24000.00	1	2024-05-10
2	Sneha	NULL	Tamang	sneha.tamang@ex.com	22000.00	2	2023-10-01
3	Bibek	Prasad	Shah	bibek.shah@ex.com	26000.00	3	2022-08-15
4	Laxmi	NULL	Malla	laxmi.malla@ex.com	20000.00	4	2025-02-20
5	Kamal	Raj	Thapa	kamal.thapa@ex.com	23000.00	3	2023-06-30
6	Nabin	NULL	Rai	nabin.ra123@ex.com	21000.00	6	2024-07-12
7	Pooja	Devi	Shrestha	pooja.shres@ex.com	25000.00	1	2022-11-05
8	Suman	NULL	Lama	suman.lama12@ex.com	19500.00	8	2023-03-18
9	Asha	Kumari	Yadav	asha.yadav12@ex.com	21500.00	9	2024-09-25
10	Manoj	NULL	Karki	manoj.karki@ex.com	27000.00	2	2025-01-01

4. Insert a new employee without their joindata.

Query: INSERT INTO employee(fname, mname, lname, Email, salary, did)
VALUES('uncle','kumar','jha','adsss@gmail.com',20000,1)

5. Insert a new employee of salary 15000.

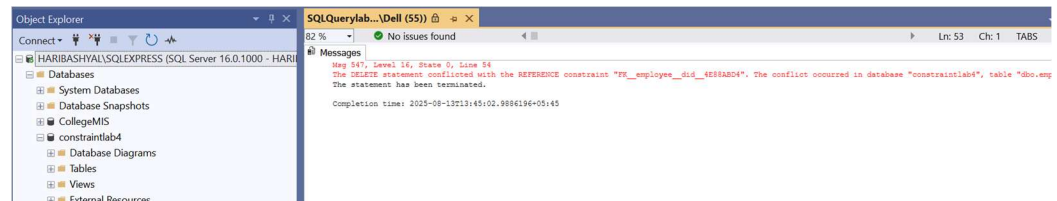
Query: INSERT INTO employee(fname, mname, lname, Email, salary, did)
VALUES('uncle','kumar','jha','adsss@gmail.com',15000,1)

6. Insert two new employee of email same.

Query: INSERT INTO employee(fname, mname, lname, Email, salary, did)
VALUES('anuj','kumar','jha','abd@gmail.com',150000,1),
('bobby','kumar','jha','abd@gmail.com',150000,1)

7. Delete IT department.

Query: DELETE from department
WHERE dname = 'IT'



8. List all employees who do not join any department.

Query: SELECT *
from employee e
LEFT JOIN department d
ON e.did = d.did
WHERE d.did IS NULL;

